

Elo-Based Evaluation of Pass Blocking and Pass Rushing Performance in the NFL

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Abstract

Quantifying pass protection and pass rushing at the player level remains a core challenge in football analytics. Common summaries such as sacks, hits, and aggregate win-rate metrics contain useful signal, but they are sparse, context-dependent, and typically unadjusted for opponent strength. Using one full NFL season of tracking data (94,539 blocker–rusher interactions across 20,980 pass plays), we develop a generalized Elo rating system that updates player ratings sequentially at the interaction level. The model combines (i) opponent-adjusted expectations through Elo’s win-probability mapping, (ii) a multi-outcome scoring rule that distinguishes `beat`, `hit`, and `sack` by severity, and (iii) an adjustment for double-team situations. In a chronological holdout evaluation, Elo achieves lower mean squared error than a baseline based on player-specific empirical outcome frequencies (0.01490 vs. 0.01493; $p = 2.76 \times 10^{-5}$). End-of-season Elo ratings also align with coefficients from a ridge-regularized Bradley–Terry model, providing convergent validity. Together, these results suggest that Elo-style dynamic ratings offer a practical, opponent-adjusted complement to existing pass-rush and pass-block evaluation metrics.

Keywords— Elo ratings, NFL, pass blocking, pass rushers, predictive modeling

1 Introduction

Pass protection and pass rushing are central to modern NFL efficiency and roster construction, but player-level evaluation in the trenches remains difficult. For rushers, box-score outcomes such as sacks and hits are rare and heavily mediated by context (e.g., quarterback time-to-throw, coverage, and game situation). For blockers, the inverse problem holds: a lineman can execute consistently without generating a direct box-score statistic.

To better isolate line play, ESPN introduced pass block win rate (PBWR), which measures whether a blocker sustains a block for 2.5 seconds on a pass play (Burke, 2018). More recent work uses tracking-based continuous measures of pressure development, such as STRAIN, which summarizes how quickly defenders close space to the quarterback over time (Matthews, 2023). These approaches provide richer information than sacks alone. However, they are typically reported as season-level aggregates and do not inherently adjust for opponent strength. In the NFL, where matchups are structured by divisions and conferences, failing to account for opponent quality can systematically bias player ratings upward (soft schedules) or downward (hard schedules).

Rating systems that adjust for opponent strength are standard in head-to-head competitive settings

(e.g., chess, esports) and in sports power ratings (Jones, 2025; Analytics, 2018). Our goal is to bring this logic to trench play by treating each pass-rush engagement as a repeated matchup between a pass rusher and a pass protector. We introduce a generalized Elo framework for NFL pass rushers (defensive linemen and linebackers) and pass blockers (offensive linemen) that updates ratings sequentially after each interaction using (i) pre-play ratings, (ii) observed engagement outcomes, and (iii) whether the rusher is double-teamed.

Unlike standard Elo, which assumes a binary win/loss outcome, we map four ordered outcomes—**block**, **beat**, **hit**, and **sack**—to a continuous score that reflects defensive impact, allowing rating updates to scale with play severity rather than treating all wins as equal. A practical challenge in this domain is strong outcome asymmetry: most pass-rush engagements end in a successful block. To encode this baseline while stabilizing early-season estimates, we initialize blockers and rushers on different rating baselines, which functions as an empirical Bayes prior reflecting that league-average blockers are expected to win most matchups.

We evaluate the resulting Elo ratings in two ways. First, we measure out-of-sample prediction error on a held-out time block and compare against a baseline based on player-specific empirical outcome frequencies. Second, we compare Elo ratings with coefficients from a regularized Bradley–Terry model estimated on the same interaction data.

Contributions. Our main contributions are:

1. A play-level Elo system for pass protection and pass rushing that adjusts for opponent strength.
2. A multi-outcome scoring rule that distinguishes the magnitude of a defensive win.
3. A simple adjustment for double-teams and a learning-rate schedule that stabilizes ratings with sample size.
4. Empirical validation via time-based out-of-sample prediction and comparison to a penalized Bradley–Terry model.

2 Methods

2.1 Data Processing

We use NFL player-tracking data from the 2021 season (regular season and postseason). Tracking coordinates are recorded at 10 Hz (every 0.1 s) and are accompanied by event annotations marking key moments such as the start of a blocking engagement, pass attempts, and sacks.

We restrict attention to dropbacks by retaining plays containing a forward pass event or a quarterback sack event. Within each retained play, we keep offensive linemen, identified pass rushers, and the quarterback. For each frame t , we compute the Euclidean distance between the quarterback and any non-quarterback player:

$$d_{p,t} = \sqrt{(x_{p,t} - x_{QB,t})^2 + (y_{p,t} - y_{QB,t})^2}. \quad (1)$$

Our unit of analysis is a *blocker–rusher interaction*, defined as an engagement between a specific pass rusher and the primary pass protector on that play, with an indicator for whether the rusher

is double-teamed.¹

We assign one outcome per interaction using severity priority **sack** > **hit** > **beat** > **block**. A **beat** occurs if the rusher gains a decisive advantage within 2.5 seconds, operationalized as the rusher becoming closer to the quarterback than the blocker before 2.5 seconds. A **sack** is taken directly from event annotations. A **hit** is recorded when the quarterback is contacted before pass release without a sack. If none of these events occurs, the interaction is labeled **block**.

After filtering and labeling, the dataset contains $n = 94,539$ blocker-rusher interactions across 20,980 pass plays in 279 games. There are 337 unique blockers and 598 unique rushers. The outcome frequencies are: sack = 1.12%, hit = 0.497%, beat = 13.2%, and block = 85.2%. In 46.2% of interactions the rusher is double-teamed.

2.2 Elo Model

Standard Elo models assume a binary outcome. We generalize Elo by mapping each interaction outcome to a continuous observed score in $[0, 1]$ from the rusher’s perspective. Intuitively, **sack** is a strong defensive win, **hit** is intermediate, **beat** is a mild win, and **block** is a loss for the rusher. We assign scores using a monotone scale informed by average play impact (e.g., EPA), rescaled to $[0, 1]$ (Eager, 2018):

$$S_A^{(t)} \in \{0, 0.2, 0.4, 1\} \quad \text{for} \quad \{\mathbf{block}, \mathbf{beat}, \mathbf{hit}, \mathbf{sack}\}, \quad S_B^{(t)} = 1 - S_A^{(t)}.$$

Let $R_A^{(t)}$ and $R_B^{(t)}$ denote the pre-play ratings for rusher A and blocker B at interaction t . The expected score for the rusher is

$$E_A^{(t)} = \frac{1}{1 + 10^{(R_B^{(t)} - R_A^{(t)})/\alpha}}, \quad E_B^{(t)} = 1 - E_A^{(t)},$$

with scale parameter $\alpha = 400$.

Double-teams. Double-teams reduce a rusher’s chance of winning the interaction but also convey information about perceived rusher quality. We incorporate double-teams by adding a temporary rating bonus of 100 to the blocker side before computing expectations. This shifts the implied win probability by approximately 10 percentage points, broadly consistent with the observed change in win rates between single- and double-team situations in our sample.

Rating updates and learning rates. Ratings are updated after each interaction as

$$R_A^{(t+1)} = R_A^{(t)} + K_{A,t} (S_A^{(t)} - E_A^{(t)}), \tag{2}$$

$$R_B^{(t+1)} = R_B^{(t)} + K_{B,t} (S_B^{(t)} - E_B^{(t)}). \tag{3}$$

We use player-specific learning rates $K_{A,t}$ and $K_{B,t}$ that decay with the number of prior interactions for that player: $K = 40$ for the first 25 interactions, then linearly decaying to $K = 20$ by 225 interactions, with a floor of $K = 20$ thereafter. Because K can differ between the two participants, the updates are not strictly zero-sum across the two rating pools; this is appropriate here because we model separate populations (blockers and rushers) with different priors and sample sizes.

¹This definition follows the engagement labeling in the tracking data. When multiple blockers are assigned to the same rusher, we retain the primary blocker for the interaction and set the double-team indicator to 1.

Initialization. Pass protection outcomes are highly asymmetric: most interactions end in a successful block. To encode this baseline, we initialize blockers at 1120 and rushers at 750, which under $\alpha = 400$ implies a prior rusher win probability of $1/(1 + 10) \approx 0.091$ for an average matchup. This acts as an empirical Bayes prior that stabilizes early-season ratings when individual sample sizes are small.

3 Results

We fit the Elo model on the full interaction dataset ($n = 94,539$). Figure 1 shows the distributions of final Elo ratings for blockers and rushers. Ratings are approximately unimodal within each group and centered near their initialization values, with a right tail corresponding to consistently high-performing players.

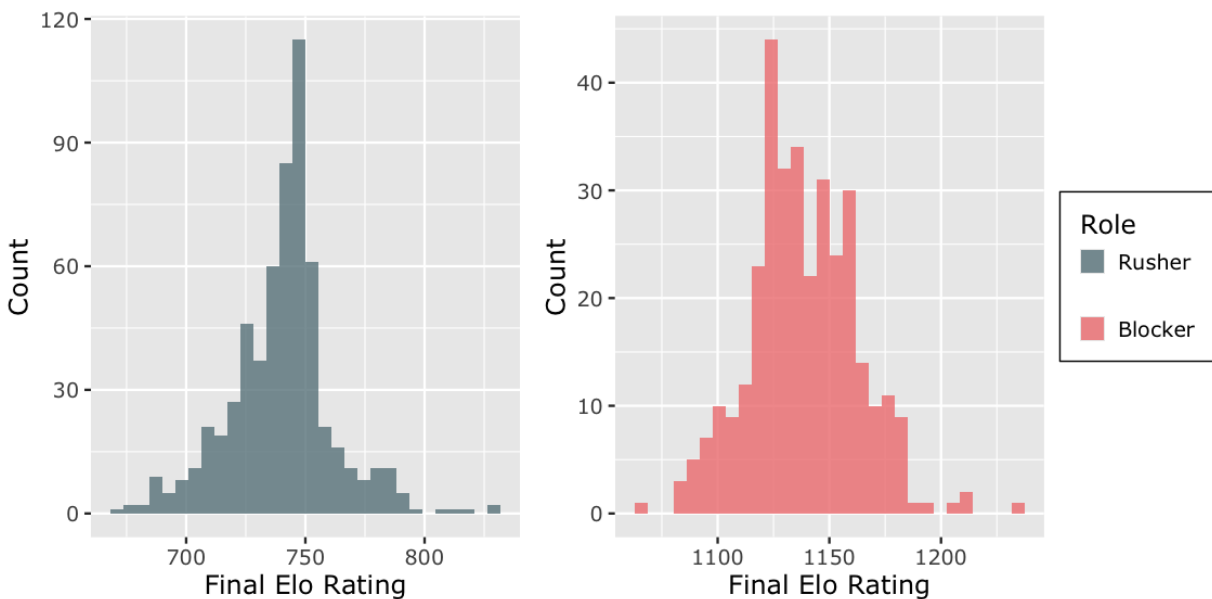


Figure 1: Elo rating distributions within each position group.

Figure 2 plots rating trajectories for the top 10 rushers and top 10 blockers (ranked by final Elo). These trajectories typically rise above baseline early in the season and then stabilize as the learning rate decays with sample size.

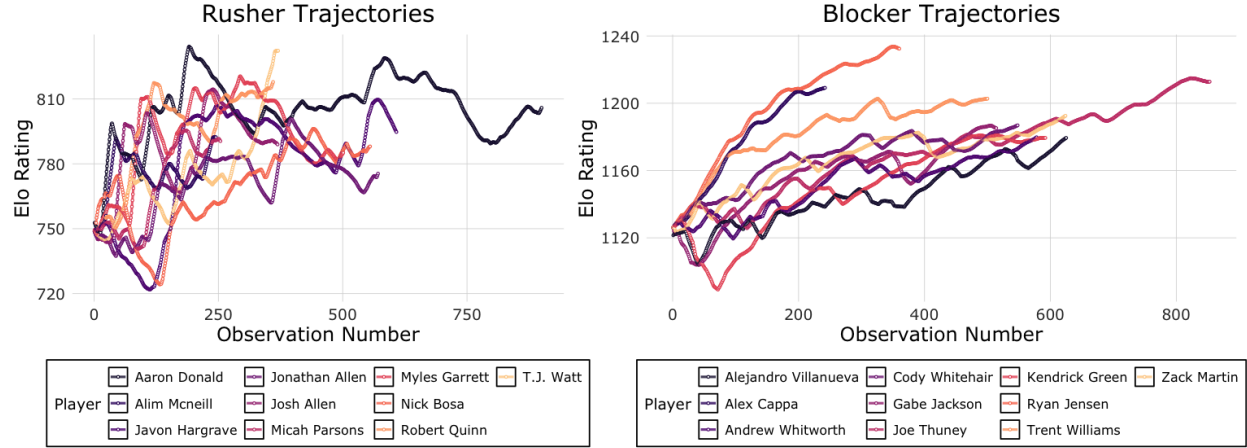


Figure 2: Top 10 rusher and blocker trajectories (ranked by final Elo), smoothed by a rolling average of width 20 to reduce noise and sensitivity.

For context, Figure 3 shows trajectories for randomly sampled players; these fluctuate around baseline with substantially smaller net drift.

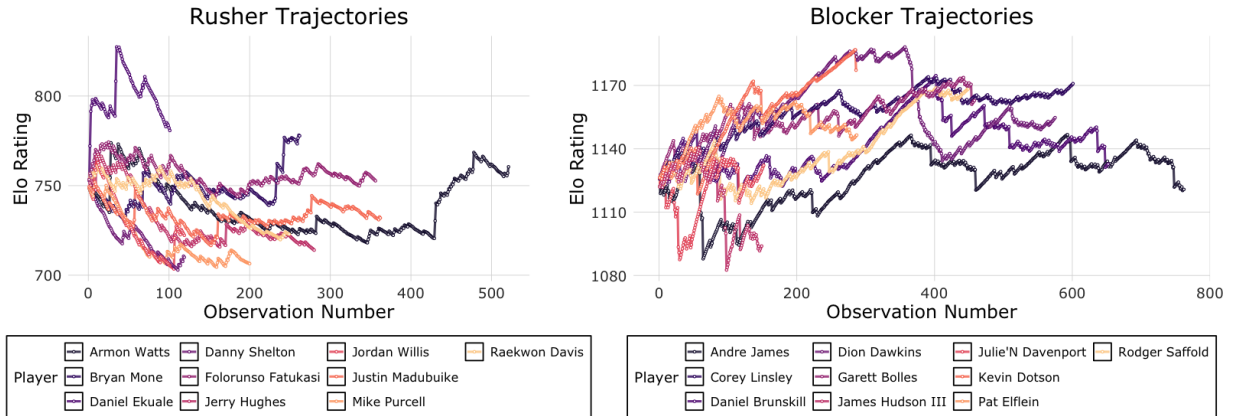


Figure 3: Randomly selected rusher and blocker trajectories.

4 Validation

4.1 Reference Model Comparison

We evaluate predictive performance on a held-out *time block* because Elo is order-dependent. The training set consists of the first 80% of interactions in chronological order; the remaining 20% form the test set.

Baseline. For each rusher, we compute training-set outcome frequencies and convert them to an expected score using the same scoring rule as Elo: sack = 1, hit = 0.4, beat = 0.2, block = 0. The baseline prediction for a test interaction is that rusher-specific expected score.

Elo prediction. For Elo, we freeze each player’s rating at its final value at the end of the training period and compute the predicted expected score E_A for each test interaction via the Elo expectation formula.

Evaluation. On the test set ($n = 18,767$ interactions with complete predictions), Elo achieves $\text{MSE} = 0.01490$ compared to 0.01493 for the baseline. Although the improvement is small in absolute terms, it is consistent across interactions. We test the per-interaction difference in *squared error* using a paired test and find the improvement is statistically significant ($t(18766) = -4.03$, $p = 2.76 \times 10^{-5}$).²

4.2 Regularized Bradley–Terry Model

As a second validation, we compare Elo ratings to ability parameters from a regularized Bradley–Terry logistic model. For each interaction $t = 1, \dots, n$, let $Y_t \in \{0, 1\}$ indicate whether the rusher achieved a win (beat/hit/sack = 1; block = 0). Let $i(t)$ index the rusher, $j(t)$ the blocker, and let D_t indicate a double-team.

We model

$$\text{logit } \Pr(Y_t = 1) = r_{i(t)} - b_{j(t)} + \delta D_t,$$

where r_i and b_j are rusher and blocker ability parameters and δ captures the marginal effect of a double-team on rusher win probability. To stabilize estimates under sparsity, we fit a ridge-penalized model:

$$\arg \min_{r, b, \delta} \left\{ -\ell(r, b, \delta) + \frac{\lambda}{2} \left(\sum_i r_i^2 + \sum_j b_j^2 + \delta^2 \right) \right\}.$$

We select λ by cross-validated predictive performance. Since Y_t is coded from the rusher’s perspective, larger r_i indicates better rushing ability, while larger b_j indicates better blocking ability (because b_j enters with a minus sign).

Figure 4 plots the relationship between Elo ratings and ridge estimates. We observe the expected monotone association: high-Elo rushers tend to have high r_i , and high-Elo blockers tend to have high b_j (equivalently, low values of $-b_j$). This agreement provides convergent validity between a dynamic rating approach (Elo) and a standard static paired-comparison model. The relationship is noisier for rushers than blockers, with a cluster of rushers near average Elo but elevated ridge estimates. One plausible explanation is that Elo uses graded outcomes (**beat/hit/sack**), whereas the Bradley–Terry model collapses outcomes to a binary win indicator.

²Because interactions within a game are correlated, we also recommend reporting a block bootstrap or clustering standard errors at the game level as a robustness check.

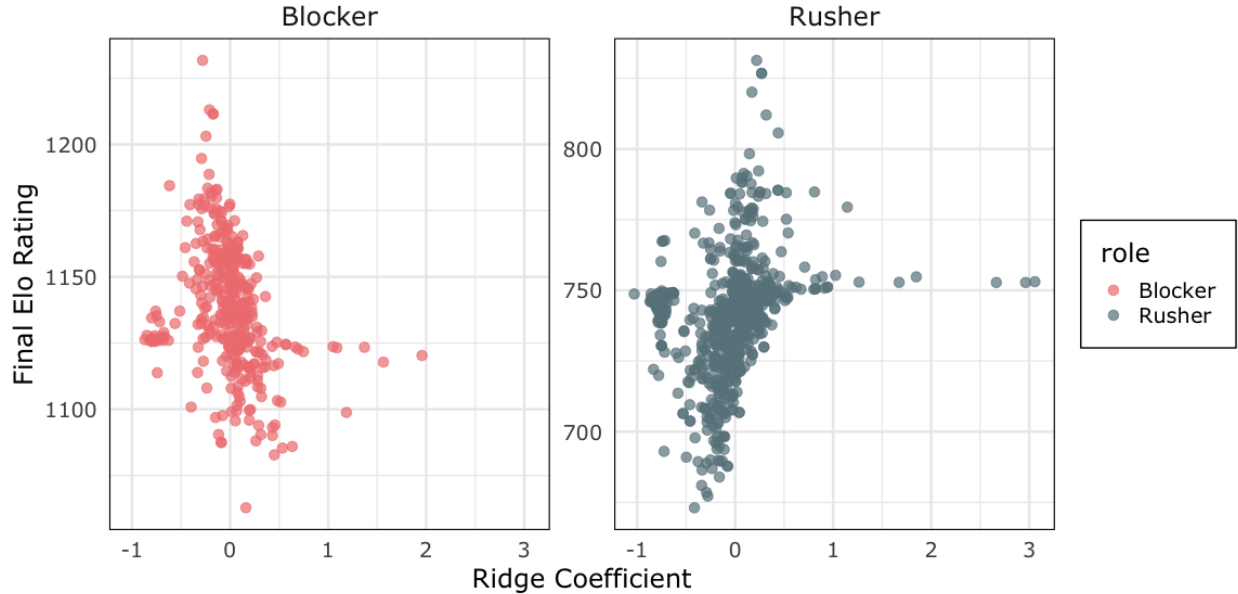


Figure 4: Final Elo rating plotted versus ridge coefficient.

Correspondence with accolades. As an external validity check, we compare end-of-season Elo ranks and ridge coefficients with All-Pro selections. Most selected players in our sample have above-baseline model values, and several appear among our top-ranked players by position group. We also observe notable exceptions (e.g., Jason Kelce and Darius Leonard), highlighting that awards and play-by-play matchup metrics capture overlapping but non-identical notions of performance.

Table 1: Selected All-Pro players with ridge coefficients and Elo ratings.

Player	Ridge	Elo	Position
Trent Williams	-0.2461683	1203.0631	Blocker
Tristan Wirfs	-0.3074198	1167.7781	Blocker
Joel Bitonio	-0.1065518	1174.8067	Blocker
Zack Martin	-0.2914964	1194.6953	Blocker
Jason Kelce	0.1719285	1123.4402	Blocker
T.J. Watt	0.2193566	831.3049	Rusher
Myles Garrett	0.1626730	787.6586	Rusher
Aaron Donald	0.3169844	812.0068	Rusher
Cameron Heyward	-0.1641323	735.5349	Rusher
Micah Parsons	0.0689751	788.2134	Rusher
Darius Leonard	-0.7197357	728.1193	Rusher
De’Vondre Campbell	0.1830619	739.6728	Rusher

5 Discussion

We propose a generalized Elo framework for evaluating pass protection and pass rushing from play-level tracking data. The model (i) supports multi-level outcomes rather than binary win/loss labels,

(ii) updates ratings dynamically over time, and (iii) adjusts expected performance for opponent strength. In held-out time-block evaluation, Elo modestly but significantly improves predictive accuracy relative to a baseline based on player-specific empirical outcome frequencies, and it is broadly consistent with ridge-regularized Bradley–Terry estimates.

These results suggest three practical advantages in this setting: opponent-adjusted ratings without requiring a fully specified hierarchical model, adaptation to non-stationary performance over the season, and sequential updates that naturally emphasize recent performance through interaction-dependent learning rates. In practice, this makes Elo a useful complement to season-aggregate metrics such as PBWR and continuous tracking measures such as STRAIN, especially for near-real-time assessment under changing matchups and roles.

5.1 Limitations and Future Directions

Several limitations suggest clear extensions. First, our **beat** labeling rule is simple and relies on distance-based heuristics. More realistic labeling could incorporate pocket geometry, the rusher’s lane relative to the quarterback’s drop, and whether pressure materially constrains the quarterback. Second, the current pipeline does not explicitly model blocks by tight ends or running backs (chips), which can affect rusher outcomes in ways correlated with opponent quality. Third, both sacks and proximity-based pressure are affected by quarterback time-to-throw. Incorporating a **hurry** label may partially alleviate this dependence by rewarding rushers who force quicker releases, but this is difficult to identify with spatial data alone. Finally, although Elo adjusts for opponent skill, it does not explicitly account for teammate effects. A center compensating for weak guard play may see his rating absorb team-level context, and vice versa.

5.2 Conclusion

Overall, Elo-based interaction ratings provide a lightweight, opponent-adjusted framework for trench evaluation and are most valuable when paired with richer contextual features in future work.

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